MIL-C-28998A(MC)
16 September 1980
SUPERSEDING
MIL-C-28998(MC)
1 June 1973

MILITARY SPECIFICATION

CLEANER-DRYER: ELECTRONIC TEST INSTRUMENT, PORTABLE

This specification is approved for use by the Marine Corps, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes requirements for one type of electronic test instrument cleaner-dryer, complete with external controls, lamps, spray system and air compressor, utilizing water and a liquid detergent cleaning compound in the cleaning process.

2. APPLICABLE DOCUMENTS

*2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-T-60	-	Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproof, Flexible.
MIL-T-152	- Treatment, Moisture-and Fungus-Resistant, Of Communications, Electronic, and Associated Electrical Equipment.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-E-52798	- Enamel, Alkyd, Camouflage.

STANDARDS

MILITARY

MIL-STD-129	- Marking For Shipment And Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking And Waterproofing; With Appropriate Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified in the contract or order (see 6.2), prior to the submission of any completed cleaner-dryers for acceptance, the contractor shall furnish a first article to determine compliance with this specification. The first article shall be identical in all respects to the cleaner-dryers which the contractor proposes to furnish as production items. Any deviations or changes in production units from the first article shall receive prior written approval from the contracting officer. The first article may be submitted as one of the production units, providing it has been refurbished and all defects noted during the first article tests have been corrected. This unit shall be subjected to the acceptance inspection.

3.2 Parts and material. Parts and materials shall be as specified herein. Parts and materials not covered by applicable specifications shall be of best commercial quality and of the lightest practicable weight entirely suitable for the purpose intended. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In all cases, the meeting of general material requirements such as nonflammability, fungus resistance, and nontoxicity shall be of prime consideration.

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3.2.1 Tropicalization. Tropicalization of the cleaner-dryer shall be effected by the selection of fungus inert parts and materials. Where the foregoing is impractical, the parts and materials shall be treated in accordance with MIL-T-152 prior to installations so that overall spraying is not necessary.

3.3 Design and construction. The cleaner-dryer shall consist of a cleaner unit, a drying oven, and a small mobile electric motor driven air compressor. The cleaner-dryer shall meet the following requirements.

*3.3.1 Power requirement. Power input requirements for the cleaner-dryer shall be 115/230 alternating current volts (Vac), ±10 percent, 60 hertz (Hz), ±5 Hz, 1-phase, 3-wire.

3.3.2 Cleaner unit. The cleaner unit shall be of welded steel construction, enclosed on all sides except the front which shall be open to allow access to the interior of the cleaner unit. The cleaner unit shall be capable of being gravity mounted on a steel mounting stand (see 3.3.4) and shall contain the following components.

*3.3.2.1 Rotary turntable. A rotary turntable shall be provided to allow positioning of equipment for the cleaning process. The turntable shall be capable of supporting equipment weighing up to 150 pounds. The surface of the turntable shall be finished with a suitable material to withstand the wear of repeated placement and removal of various electrical, electronic and mechanical items.

3.3.2.2 Lighting. A multiple floodlamp lighting system shall be provided; sufficient illumination shall be furnished by this lighting system to allow inspection of equipment being cleaned without the use of additional lamps or lighting systems. Devices to protect floodlamps from cold water spray shall be provided.

3.3.2.3 Exhaust blower. An exhaust blower shall be provided to allow exhausting of fumes and mists generated during the cleaning process.

3.3.2.4 Cleaning solution tank. A 1-gallon capacity cleaning solution tank of unbreakable material shall be provided. It shall be capable of being filled easily and shall be removable for cleaning.

*3.3.2.5 Spray cleaning and air guns. Separate spray cleaning-rinsing and air guns, each capable of independent operation, shall be provided. The rate of flow of the cleaning solution shall be nominally

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11 to 13 ounces of liquid per minute with a minimum of 50 pounds force per square inch (lb_f/in^2) of pressure supplied by the air source. The rate of flow of clear rinse water shall not exceed 32 ounces per minute. The spray cleaning gun shall have a pistol grip configuration with on-off capability. The air gun shall be a common straight through type with on-off valve.

3.3.2.6 Water pump. An electrically driven water pump having a nominal delivery rate of 2 pints per minute at a minimum water pressure of 30 pounds per square inch at the inlet to the cleaner unit shall be provided. This delivery rate and water pressure shall be provided when the water pump is located up to a maximum of 100 feet from the cleaner unit. The water pump outlet shall be connected to the cleaner unit inlet by rubber hoses of a size to allow the required delivery rate and water pressure. The hose connectors adapters shall mate with the cleaner unit's inlet and the water pump outlet. A short length of rubber hose shall be used to connect the water pump inlet and a 400-gallon capacity military water trailer having a female 3/4-inch-14 NPT outlet. The water pump, hoses, fittings, adapters, and connections shall be easily removable using common hand tools and shall be configured to allow compact stowage within a shipping crate (see 5.2). The following items shall be provided as accessories to the water pump:

(a. Four 25-foot lengths of 3/4-inch inside diameter (id) reinforced rubber hose to provide connection between the water pump outlet and the cleaner unit's inlet.

(b. One 10-foot length of 3/4-inch id reinforced rubber hose to provide connection between the water pump inlet and the military water trailer.

(c. Necessary quantity of adapters, fittings, or connectors to allow connection between the water pump and the cleaner unit and the water pump and the military water trailer.

(d. One 100-foot length of electrical power cable to provide electrical power to the water pump motor.

3.3.2.7 Fittings, connections, and hoses. A series of fittings, connections, and hoses shall be provided as follows for connection of domestic water supply or water pump, compressed air supply and drainage.

3.3.2.7.1 Exterior fittings and connections. The inlet water fitting shall allow connection of a domestic water supply or the water pump through a common garden hose. The compressed air inlet fitting shall allow connection of the air supply through a reinforced rubber hose capable of withstanding a minimum pressure of 200 lb_f/in^2. The drainage fitting

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shall be located to provide for good drainage of the cleaner unit and shall allow connection of a common garden hose. All fittings and connectors be easily replaceable.

3.3.2.7.2 Interior fittings, connections and hoses. Interior fittings, connections, and hoses shall allow connection between the inlet water supply, cleaning solution tank and spray cleaning gun and connection from the compressed air inlet fitting to the air gun. Interior hoses shall be of rubber or reinforced rubber entirely suitable for the purpose intended.

3.3.2.8 Control panel. A control panel mounted on the top forward edge of the cleaner unit shall be provided. It shall contain an ON-OFF switch, water pump switch, automatic valving system capable of providing FILL, CLEAN, and RINSE cycles, an adjustable air pressure regulator and a gage adjustable from 0 to 100 pounds per square inch. A fuse shall be provided for protection of lighting and exhaust blower motor.

3.3.3 Drying oven. The drying oven shall be of welded steel construction. It shall be enclosed on all sides except the front which shall be contain one or two steel doors hinged at the sides and having a latching device to retain the door in an open or closed position. Two interior perforated steel shelves capable of supporting 100 pounds each shall be provided. These shelves shall be adjustable to allow positioning of the shelves to accommodate various sizes of equipment or a number of small items. The drying oven shall be capable of being gravity mounted on a steel mounting stand (3.3.4). The drying oven shall meet the following requirements.

*3.3.3.1 Drying and operation. The drying oven shall be a low temperature oven and shall be designed to provide for the safe drying of electronic instruments. It shall contain no thermostatic elements or contactors which would "hang-up" or otherwise cause a potentially damaging over-temperature condition. The temperature control shall be inherently "fail safe" and shall allow unattended and continuous operation.

*3.3.3.2 Temperature control. The drying oven shall have a control box mounted on the exterior front top. The control box shall contain a fuse capable of handling the input power specified in 3.3.1, an on-Off switch, a power-on indicator lamp and a variable temperature control (rheostat) which shall be capable of varying the input voltage to the infrared lamps, thereby controlling the oven temperature from ambient to +52°C (+125°F). Maximum put power shall be 1,500 watts.

3.3.4 Mounting stands. Mounting stands for the cleaner unit and

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the drying oven shall be provided. They shall be of welded steel construction and shall be capable of supporting the cleaner unit or drying oven with 200 pounds of equipment placed therein. The stands shall be of a height to place the working surface of the rotary turntable (see 3.3.2.1) of the cleaner unit and the lower shelf of the drying oven, at a height of 30 inches above ground level. Leveling adjustments for the mounting stands shall be provided.

3.3.5 Compressed air supply. A mobile electric motor driven air compressor with steel storage tank shall be provided, capable of delivering 3.5 cubic feet per minute of air at 50 lb_f/in^2, utilizing a reinforced rubber air hose of a size to allow the required air delivery. The electric motor and air compressor shall be mounted on a steel platform which is affixed to the top of the compressed air storage tank. The steel platform shall contain adjustment slots to allow for the positioning of the electric motor to increase or decrease tension on compressor drive belt(s). A belt guard shall be provided as a safety feature. The storage tank shall have rubber tired wheels and a steel handle bar for easy mobility.

*3.3.5.1 Electric motor. The electric motor shall have a nominal rating of 1 horsepower and operate from a 115/230 Vac, 60 Hz, single phase electrical source. It shall have overload protection, automatic reset capability and adequate protection against adverse weather conditions.

*3.3.5.2 Air compressor. The air compressor shall have adequate capacity to provide 5 cubic feet per minute at 50 lb_f/in^2 (see 3.3.5) and shall be belt driven by the electric motor. It shall be attached to the steel platform by bolts suitable for the purpose intended.

*3.3.5.3 Air storage tank. The compressed air storage tank shall be of welded steel construction and shall have a storage capacity of 20 gallons. Safety devices shall be provided to prevent over-pressurization of the storage tank. A pressure switch which will stop the air compressor at 100 lb_f/in^2 and restart the air compressor at 80 lb_f/in^2 shall be provided. The storage tank shall have a valve or petcock to allow "bleed-off" of air pressure and drainage of condensation and a pressure gage to indicate air pressure within the storage tank.

*3.3.6 Safety devices. The cleaner-dryer shall have the necessary safety devices or features to protect operating personnel from electrical shock, safety goggles to protect the operator's eyes from contaminants, air compressor belt guard to prevent entanglement of clothing or injury to the operators limbs and devices to protect the operator from burns caused by accidental touching of the cleaner unit floodlamps or the drying oven infrared lamps. A safety pressure relief valve to prevent over-pressurization of the air storage tank shall be provided.

3.4 Dimensions. The dimensions of the cleaner-dryer shall be as follows:

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(a) Cleaner unit. The cleaner unit without the water pump shall have the following overall exterior dimensions which shall encompass all protrusions.

Width------32 -0 +1/2 inches

Depth------32 -0 +1/2 inches

Height-----47 -0 +1/2 inches

(b) Drying oven. The drying oven shall have the following overall exterior dimensions which shall encompass all protrusions:

Width------34-1/2 -0 +1/2 inches

Depth------23 -0 +1/2 inches

Height-----46 -0 +1/2 inches

(c) Mounting stands. The mounting stands shall have the approximate width and depth of the cleaner unit and the drying oven; the height shall be determined by the requirements of 3.3.4.

*3.5 Weight. The weight of the cleaner-dryer shall be as follows:

(a) Cleaner unit. The weight of the cleaner unit without water pump shall not exceed 250 pounds with all components installed.

(b) Drying oven. The weight of the drying oven shall not exceed 185 pounds with all components installed.

(c) Mounting stands. The weight of each mounting stand shall not exceed 35 pounds.

*3.6 Cleaning, treatment and painting. The exterior surfaces of the the cleaner-dryer shall be cleaned, treated and painted as specified in MIL-F-704, type A. The finished coat shall be in accordance with MIL-E-52798 (forest green).

3.7 Marking. Nameplates of durable material shall be attached to, and identify, each major component of the cleaner-dryer. Nameplates shall also contain the name of the manufacturer and the contract number. The various positions of switches, valves, etc., shall be identified. Marking shall be permanent and of best commercial practice.

3.8 Workmanship. The cleaner-dryer, including all parts and assemblies, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects,

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burrs, and sharp edges; accuracy of dimensions, marking of parts and assemblies; thoroughness of soldering, welding, brazing, wiring and riveting; alignment and mating of parts and assemblies and tightness of assembly screws and bolts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3).

(b) Acceptance (see 4.4).

(c) Preparation for delivery (see 4.7).

4.3 First article inspection. The first article shall be subjected to all inspections and tests specified in 4.6 and 4.7. The first article inspection and testing shall be jointly witnessed by technical representatives of the contracting officer and cognizant DSAR. Approval of the first article will be in writing from the contracting officer.

4.4 Acceptance inspection. Each cleaner-dryer submitted for acceptance under the contract shall be subjected to the inspection specified in 4.6.1 and the test in 4.6.2.

4.5 Rejected units. Any cleaner-dryer which does not meet the full requirements of this specification shall be rejected. Rejected units shall be suitably marked to indicate cause, giving full particulars of previous rejections and action taken to prevent recurrence.

4.6 Tests. Inspections and tests shall be conducted at room temperature and the test fluid shall be water. All inspection and testing shall be performed in the manner specified herein. Any damage, deterioration or change in any component which could in any manner prevent the equipment from meeting performance, maintenance or service requirements during service life (unless excepted herein) shall be considered as cause for failure to pass the particular test or inspection.

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4.6.1 Visual and mechanical inspection. The cleaner-dryer, including its components, assemblies and parts, shall be given a thorough visual and mechanical inspection to determine that the quality of materials and workmanship is in compliance with the requirements of this specification and all referenced documents. Particular attention shall be given to the following:

(a) Completeness

(b) Nameplates, identification markings and labels.

(c) Ease of operation of adjustable, sliding and rotating parts, doors, shelves, and ventilator coverings.

(d) Finishes

(e) Weldments

(f) Lubrication and rust prevention

(g) Fastening and securing devices or parts

(h) Accessibility of components and parts for servicing and use

(i) Overall dimensions

(j) Weight

(k) Burrs and sharp edges

(l) Foreign objects, chips and dust

(m) Electrical wiring, fittings, connections, and outlets

(n) Mechanical gaskets, seals, fittings, connectors, and drains

(o) Rubber hoses, clamps, and adapters

(p) Damage survey

(q) Interchangeability

4.6.2 Operation test. The cleaner-dryer shall be operated to determine compliance with the requirements herein and to determine proper operation of the air compressor motor, air compressor, air pressure regulators, pressure relief valves, water pump selector valves, solenoid valves, turntable rotation,

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light level within the cleaning unit, oven temperature, temperature controller, exhaust blower, spray and air guns, and connectors. Failure of the cleaner unit or drying oven, compressed air supply or any components, assembly or part to function properly shall be cause for rejection. During operation, the cleaner-dryer shall be inspected for leaks. Evidence of leakage shall be cause for rejection. After operation of the cleaner-dryer, all accessories, parts, hoses, and cord stowage shall be exhibited. Failure of proper stowage shall be cause for rejection.

*4.6.3 Endurance test. The cleaner-dryer shall be subjected to 24 hours of continuous operation. During this period, the cleaner unit's selector valve shall be continuously cycled through all its positions, the valve remaining in each position for 1 minute prior to cycling to the next positon. The drying oven shall be brought up to an operating temperature of +52°C (+125°F) and continuously monitored to assure that the oven temperature does not exceed +66°C (+150°F). The air compressor shall be continuously operated during the 24 hours to assure that proper air pressure is delivered to the cleaner unit. During the 24 hours of operation, the test fluid shall consist of water only, detergent solution need not be used. Minor adjustments may be made using common hand tools except that no adjustment shall be made on the controls (switches, valves, regulators, etc.). All adjustments and reasons therefore shall be recorded in the test report. No parts, components, assemblies or subassemblies shall be replaced, reworked or repaired during the 24-hour test period. Upon completion of the 24-hour endurance test, the cleaner-dryer shall be left standing for a period of 96 hours. Upon completion of the 96-hour period, the cleaner-dryer shall be visually inspected to ensure that rusting or corrosion has not occurred.

4.7 Preparation for delivery inspection. The preservation, packaging, packing and marking shall be inspected to determine compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each cleaner-dryer shall be preserved and packaged in accordance with MIL-P-116, method I. All exposed bare ferrous metal surfaces shall be coated with preservative conforming to type P-2 of MIL-P-116, covered with barrier material conforming to grade A of MIL-B-121, and secured with tape conforming to PPP-T-60. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186.

5.1.2 Level C. The cleaner-dryer shall be preserved in accordance with the suppliers commercial practice in a manner which will afford adequate protection during direct shipment to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. Each cleaner-dryer, preserved and packaged as specified in 5.1.1, shall be packed in an overseas type container conforming to PPP-B-601, style A or B. Closure and strapping shall be in accordance with the appendix of the specification. Containers shall be of uniform size and shape and of minimum tare and cube.

5.2.2 Level B. Level B packing shall be the same as level A except the containers shall be domestic type and conform to any style of the specification.

5.2.3 Level C. Each cleaner-dryer, preserved and packaged as specified in 5.1.2, shall be packed in containers which will insure safe transportation from the supplier to the frist receiving activity. Containers shall conform to carrier rules and regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cleaner-dryer covered by this specification is intended for washing, rinsing, and drying electrical, electronic, and mechanical test instruments.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specificaton.

(b) When first article is not required (see 3.1).

(c) Level of preservation, packaging, and packing (see 5.1 and 5.2).

6.2.2 Contract data requirements. Data conforming to Data Item Descriptions DI-E-4542, DI-R-4803, DI-T-4901 and DI-T-4902 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

*6.3 First article. The first article shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity.

*6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as

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a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE MIL-C-2[illegible]98A(MC)
Cleaner-Dryer: Electronic Test Instrument, Portable

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

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